POKHARA UNIVERSITY

Level: Bachelor | Semester:Spring | Year : 2024
Programme: BE | Full Marks:100
Course: Computer Architecture | Pass Marks: 45
Time :3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1.	a)	Discuss the term Computer Architecture. Explain IAS computer architecture with detailed diagram.	7
	b)	Briefly explain the terms instruction set, addressing modes and instruction formats.	8
2.	a)	What is register transfer? Briefly explain arithmetic and Logical micro-operation.	7
	b)	Explain about the internal structure of processor also explain the design principle for modern system.	8
3.	a)	Describe the steps involved in Booth's algorithm and provide an example to multiply unsigned binary number.	8
	b)	What are the significances of encoded control words in vertical microprogrammed control unit? Differentiate between hardwired and microprogrammed control unit.	7
4.	a)	Explain direct mapping and associative mapping in cache memory with necessary diagrams.	8
	b)	What is memory hierarchy. Explain with reason why do we need multilevel memory hierarchy?	7
5.	a)	What are the drawbacks of Programmed I/O and interrupt driven I/O? Explain how DMA overcomes these drawbacks.	7
	b)	Prove that "Speedup factor for pipelined processor is equal to number of stages in pipeline." Assume that pipeline has K = 6 segment and execute n = 120 tasks in a sequence. Let the time taken to process the sub-operation in each segment is 30ns. Calculate the speedup ratio.	8
6.	a)	Describe in brief about Flynn's Taxonomy. Also explain how parallelism can be experienced in uniprocessor systems.	8
	b)	Differentiate dual core and quad core processor. What are the software performance issue that arise in multicore architecture?	7

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7. Write short notes on: (Any two)
   a) BCD Adder
   b) VHDL and its advantages
   c) Array Processors

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